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FINAL PROJECT REPORT 2022/2023

Title: ASSESSING THE VULNERABILITY OF CATCHMENTS DURING FLASH FLOOD EVENTS IN SOUTHWEST ENGLAND USING MACHINE LEARNING METHODS

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Word Count:

7796 (excluding preliminary pages, appendices, and
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Abstract

As climate change in recent years continues to cause extreme weather events, the necessity for safety-oriented predictive tools has become more prevalent than ever. This project is a multidisciplinary exploration which included the development of a tool using the programming language Python. SOPHIE was developed to provide a numerical evaluation for catchment vulnerability in the southwest England region to flash flooding. The program works by determining if the catchment is a flashy catchment which indicates quick responsiveness to precipitation. Once that condition is met SOPHIE takes the North Atlantic Oscillation (NAO) and Atlantic Multidecadal Oscillation (AMO) trained machine learning model to predict the probability of a flash flood event occurring in a specific catchment. Using SOPHIE, three case studies were tested: two flash flood events and one non-flash flood event, Boscastle 2004, Totterdown 2008 and Throop 2002, respectively. All of which SOPHIE predicted to be safe as either the specific catchment was not flashy, or the probability of flash flood occurrence was below 50%. This study investigates the relationship between NAO and AMO climate indices and their predictive power for flash floods. The results indicate that the NAO and AMO relationship is an effective tool for predicting rainfall but not flash floods. Hence, other parameters should be considered for flash flood predictions. Additionally, the performance of three classification algorithms: Multi-layer Perceptron (MLP), Support Vector Machine (SVM), and K-Nearest Neighbours (kNN), are compared on a binary classification task using two datasets. The MLP algorithm outperformed the other two algorithms in terms of classification accuracy. These findings have implications for climate and weather forecasting, as well as machine learning applications in environmental studies.

Keywords: Atlantic Multidecadal Oscillation, Catchment, Flashy catchment, K-Nearest Neighbours, Machine Learning, Multi-layer Perceptron, North Atlantic Oscillation, Python, Support Vector Machine, Vulnerability

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List of Abbreviations and Mathematical Symbols

AMO	Atlantic Multi-decadal Oscillation
ANN	Artificial Neural Network
API	Application Programming Interface
NAO	North Atlantic Oscillation
NAO/AMO	NAO and AMO
NRFA	National River Flow Archive
Q_p	Peak flow
SW	Southwest
T_b	Base time
T_p	Time to peak

Acknowledgements

I would like to thank several people to whom this work would not have been possible without.

First and foremost, my academic supervisor Dr Ioanna Stamatakis a specialist in the field of water informatics with a focus on flash flood interactions with urban settlements. I thank her for her support, our meetings together, her reading the preliminary versions of this text and providing timely and informative feedback. Dr Stamatakis was also one of the few people who supported and encouraged the idea of a multidisciplinary engineering project with a programming element and for that I am extremely grateful.

I am grateful for Dr Stamatakis for connecting me with Dr Andy Barnes. He further reinforced the idea of a multidisciplinary project as he holds a PhD with the focus of machine learning in the analysis of extreme precipitation patterns. His findings were pivotal in the curation of this project, and I would like to thank him for his aid in the programming and meteorological component of this project. His contribution to this project was completely voluntary and very valuable.

Very deep gratitude goes to my Mum, Dad, little sister, little brother, and girlfriend for the constant support and understanding of my personal needs during this extremely busy time and words of encouragement. Without them this would not have been possible.

CHAPTER ONE

INTRODUCTION

1.1 Background of Project

There have been growing concerns about the global risk climate change poses, mainly relating to its ramification of catastrophic weather events occurring at a greater frequency than ever before. Of the 223 floods that have occurred worldwide in 2021 (Alves, 2023), flash floods have since become four times more likely to occur compared to the 1970s (British Red Cross, 2023). Specific land features are important when considering the effect flooding can have on the surrounding environment. This project focuses on catchments as they directly influence the severity of flash floods in several ways. The size, shape, and slope of the catchment can affect how quickly water is collected and how fast it flows downstream. This project intends to expand on the investigation of catchments during flash floods through several ways. Firstly, through the development of a tool which supports engineers in assessing the vulnerability of catchments to flash floods. Secondly, through exploring the relationship between two climate indices and flash flood events.

1.1.1 Catchments

A catchment is a basin shaped land surrounded by landforms and natural features that includes elevated land, mountains, and hills (Waikato Regional Council, n.d.). The determining feature of a catchment is that all water within a catchment drains to one point (see Figure 1).

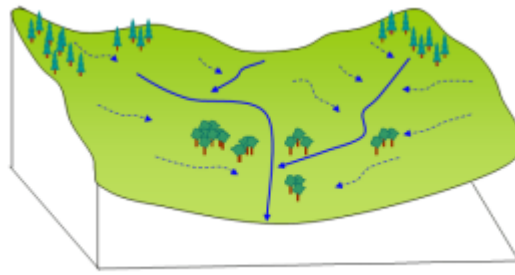


Figure 1: Diagram of catchment (Floodsite Project, 2008)

Catchments are the earth's natural response to increased rainfall; it is a drainage system. However, the danger this poses to the surrounding area is that heavy rainfall or external water sources can cause the catchment to overfill resulting in increased water levels or flooding. This is because the rate at which water is entering the catchment is increasing and the drainage rate remains the same.

1.1.2 Southwest England and flash floods

Southwest (SW) England is predominantly prone to flash floods during summer. This geographical region has been selected as the focus for this project as the University of Greenwich has conducted comprehensive research into catchments in SW England and curated a dataset featuring catchment characteristics which is all pivotal to the construction of this project's tool. The National River Flow Archive (NRFA) , a leading organisation in water management and hydrological research in the UK (NRFA, 2023), split the UK into 239 hydrometric zones. The Southwest is under hydrometric zone 43 to 53 (see Figure 2).

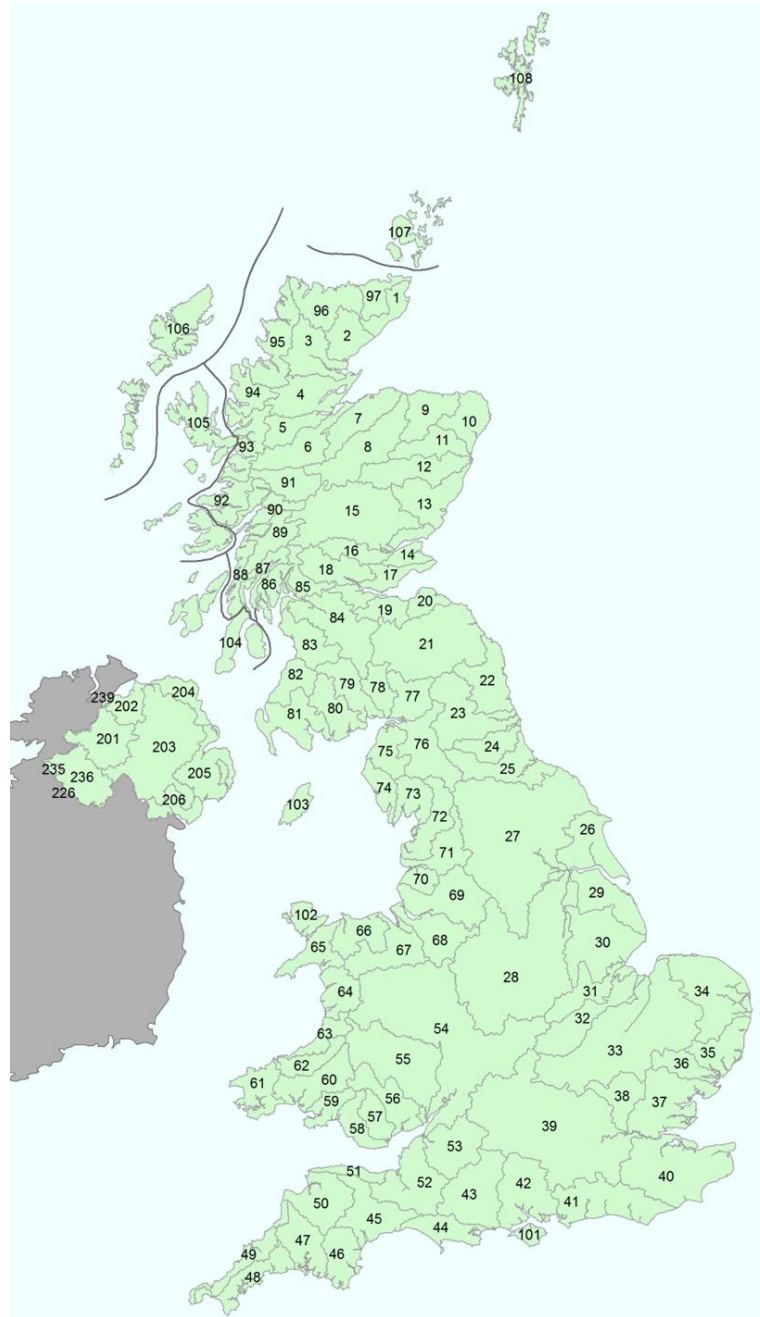


Figure 2: All hydrometric zones in the UK (NRFA, 2018)

A catchment station is a location on a river where measurements are taken to monitor a variety of hydrological parameters. The first two to three numbers of a catchment

station refer to the hydrometric zone the catchment station is located. For example, station 39056 is located in hydrometric area 39.

Majority of academic papers conclude that flash floods begin and end in the space of six hours, although there is still a level of subjectivity around the matter (Kobiyama & Goerl, 2007). The time a flood takes to start and finish is called the base time (T_b) a determining characteristic of flash floods used throughout this project is a T_b of six hours or less.

When considering the vulnerability of a catchment it is important to take the worst case scenario, in the case of flash floods this is when maximum rainfall is present. During a flood, the precipitation will reach the peak flow (Q_p), the time it takes is called the time to peak (T_p) (Langridge et al., 2020) and in this project is approximated to be half of the T_b , 3 hours.

1.1.3 Flashy catchments and basic topography

Catchments that respond rapidly to floods tend to be more vertically inclined with a larger area (Environmental Agency, n.d.). The soil in the surrounding areas tends to be highly saturated due to the increased water intake, making it difficult to effectively infiltrate more water. These types of catchments are called flashy catchments as the water levels in rivers and streams can rise quickly in response to even small amounts of rainfall and then recede just as quickly (Oxford University Press, 2022). Topographical features associated with flashy catchment include steep slopes, little vegetation cover and low infiltration rates.

1.1.4 Climate change and meteorological considerations

Climate change has been a catalyst for extreme weather events such as flooding. To investigate the vulnerability of flooding, it's necessary to consider different weather variables to predict increased rainfall level. The Icelandic Low and Azores High atmospheric pressure variation is known as the North Atlantic Oscillation (NAO) (Hurrell, 2003). The Met Office found that local pressure patterns had a significant impact on changes in weather patterns (wind, rainfall, and temperature) (Met Office, n.d.). A smaller pressure difference between the two zones can be seen during the negative NAO phase. When it comes to rainfall, a negative phase results in fewer storms (reduced rainfall). The positive NAO phase shows a greater pressure differential between the two areas, leading to more storms and warmer air from the west of the area, which could indicate more rain. The North Atlantic Ocean's cyclical variations in sea surface temperature are quantified by the AMO index. The association between the AMO characteristic patterns and the weather in various places has been found (Dutton, 2021) and can be used to identify rainfall. Similar to this, the AMO oscillates between positive and negative phases with warmer and colder water temperatures in the North Atlantic. AMO readings that are positive also signal summertime in Asia and Europe.

1.2 Aims and Objectives of the Project

1.2.1 Project Aims

The aim of this project is to develop a tool which assesses catchment vulnerability to flash flooding in SW England using topographical, hydrological, and meteorological aspects, establishing if NAO and AMO phases can be used to predict flash flooding.

1.2.2 Project Objectives

- 1 Create a tool using the Python programming language to calculate the T_p of catchments based on the catchment station characteristics, determining a catchment flashy or not
- 2 Use three machine learning algorithms to analyse the relationship between NAO/AMO and flash flooding occurrence
- 3 Use three machine learning algorithms to output if a flash flood would occur, the probability of occurrence and the accuracy of the algorithm
- 4 Make a prediction model that determines the most accurate algorithm and predicts flash flooding
- 5 Make a conclusion on whether NAO/AMO can be used to accurately predict flash floods

1.3 Definition of relevant terms

Important terms are defined in Table 1.

Table 1: Definition of relevant terms

Term	Definition
Atlantic Multidecadal Oscillation	Changes in the sea surface temperature in the North Atlantic Ocean (Dutton, 2021)
Base time	The base time is the amount of time that it takes for a river or stream to return to its normal base flow after a precipitation event
Catchment	Terrain with a basin shape and a single drainage point (Waikato Regional Council, n.d.)
Catchment station	A catchment station is a location on a river where measurements are taken of hydrological parameters

Table 1 (continued).

Flashy catchments	A catchment where the water level immediately rises following a heavy rainfall (Oxford University Press, 2022)
Flash flood	Flood starts and ends in six hours
Nation River Flow Archive	National River Flow Archive (NRFA) a leading organisation in water management and hydrological research in the UK (NRFA.2023)
North Atlantic Oscillation	Regional pressure variations in the North Atlantic (Hurrell, 2003)
Time to peak	Time when the maximum amount of rainfall is present

1.4 Scope of the Project

This project inspects multiple branches of engineering. The scope of the project needs to be predetermined so that the project has set goals, objectives and limitations. The scope can be described as a limit, this project has three main limits. First this project is limited to hydrometric areas 43 to 53 (SW England) and all models are built using catchment stations in this region (see section 1.1.2). The second limit is that all flash flood events are taken from 4th of December 1960 to 22nd of November 2016. This limit was set to match the data found for the NAO/AMO climate index. Lastly, the machine learning model complexity. Programming and simple machine learning algorithms were utilised to gain insights to an engineering exploration.

1.5 Report Outline

This report is divided into six chapters. Chapter 1 is an introduction which provides context for research and presents the project aims and objectives. Chapter 2 is a literature review which covers the topics of hydrology and flash flooding, meteorology

and climate change, topography with catchments and modelling in engineering. Chapter 3 provides the research methodology and a comparison of machine learning algorithms. Chapter 4 is the experimental results which displays the tool being tested with three different case studies and justifies the machine learning algorithm used. Lastly, Chapter 5 is the conclusion of the report which highlights key findings from the prior chapter, also future works are presented here.

Chapter 1 Introduction

Chapter 2 Literature review

Chapter 3 Research methodology

Chapter 4 Results and findings with relevant discussion

Chapter 5 Conclusion

CHAPTER TWO

LITERATURE REVIEW

2.1 Literature Review

There are 4 main themes discussed in this literature review: hydrology, meteorology, topography, and modelling in relation to precipitation.

2.2 Hydrology and flash flood events

To investigate the susceptibility of flash floods in catchments, it is important to understand the distinction between conventional floods and flash floods. Archer & Fowler (2015) discussed the analysis of historical flash floods using newspaper archives and recent gauged rainfall and streamflow data. They found five key examples in the UK to explain different types of flash floods. For each flood information about the flow rate and the max rise was collected in 15-minute intervals. A key discovery from this journal was that 'flash floods' are 'characterised by rapid rates of rise' of water levels (Archer & Fowler, 2015). Regarding catchment vulnerability it was concluded that 'steeply rising wavefronts seem to occur more commonly on steep upland catchments' (Archer & Fowler, 2015). This suggests that steepness of slope in a catchment is a topographic feature common within vulnerable catchments introducing a topographic element when considering vulnerability. Incidences of flooding can occur in catchments with less sloping given intense rainfall. Hence, weather-based data needs to be considered as well. Clark & Arellano (2004) documents the UK flood and storm events with the focus being the Louth (east midlands) storm. In May 1920 in just 20 minutes, 23 individuals were killed by the flood, and 1000 families lost their homes.

The primary cause of this was the rapid increase in peak water levels 4.6 metres in 15 minutes. More examples considered include Wansbeck catchment the rise of 30 mm in 15 minutes and Alston had 26.2 mm (Archer & Fowler, 2015). This highlights that flash floods have an extremely quick T_p value. Both examples discussed had relatively low sloping, further highlighting the need for research into weather parameters and data that lead to torrential rainfall.

Archer et al., 2019 provides insight into the significance of studying historical flash floods caused by intense rainfall, which can aid in comprehending the impact of low-frequency climate variability on flash flood frequencies. The research establishes an online flood chronology for the north and south-west regions of England that encompasses surface water and fluvial flooding, utilising local newspaper reports as the principal source of documentary material. The online chronology created was available as an online SQL database at Ceg-fepsys.ncl.ac.uk but is no longer available for public use (see Figure 3).

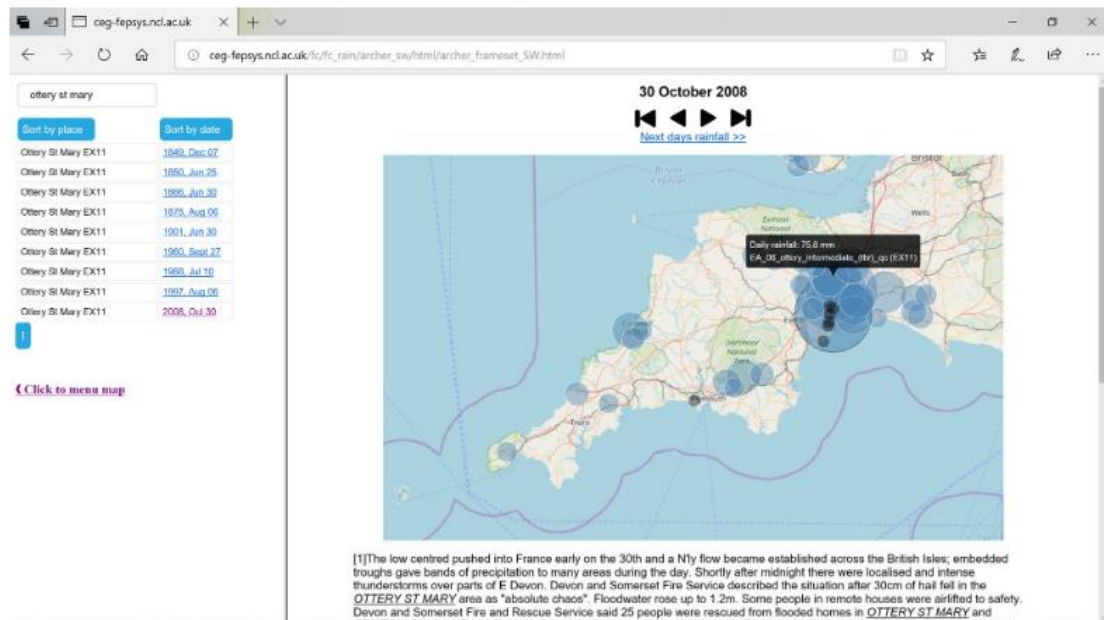


Figure 3: Archer et al., 2019 database for historical flash floods in the UK

Nonetheless the findings of the flood chronology present a useful resource for identifying catchment or settlement vulnerability to flash flood events. The findings were essential to the development of the dataset of flash floods events in the SW region, used for the tool. The figures and tables gave an approximate guideline to the amount of flash floods expected each year (see Figure 4).

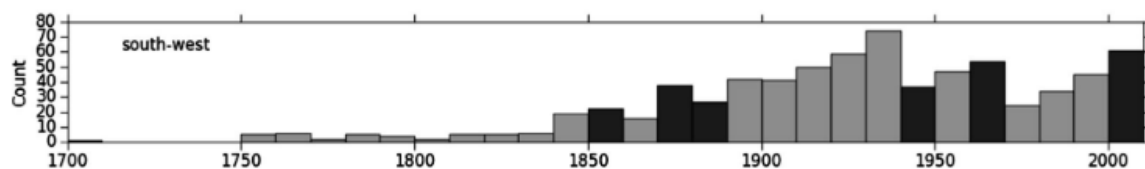


Figure 4: Number of flash flood entries per decade in SW region (Archer et al., 2019)

The paper also provided a structure for data collection and showcased acceptable reliable sources to create a dataset: this includes the use of newspaper articles, online flood chronologies and other secondary data.

2.3 Meteorology and climate indexes

Meteorology is the study of the atmosphere which involves climate and weather (National Geographic, n.d.). As introduced in section 2.2 information about weather conditions will be taken into account. There are multiple meteorological factors which can be indicative of the type of weather such as temperature, pressure changes, cloud movement, wind, and cloud spatial patterns.

The National Weather Service is an agency in the United States responsible for giving weather forecasts, warnings of dangerous weather and other weather-related products. This agency's climate and weather linkage project aims to 'monitor, assess and predict climate phenomena and their linkage to weather events' (National Weather Service, n.d.). Popular phenomena included Pacific/North American teleconnection pattern (PNA) It is characterised by changes in atmospheric pressure and circulation patterns over the Pacific Ocean and North America, which can have significant impacts on weather patterns across the region (North Carolina Climate Office, n.d.). Additionally Madden-Julian Oscillation (MJO) is a weather pattern that affects tropical regions of the Earth. It is characterised by a large-scale pattern of enhanced rainfall and atmospheric circulation that moves eastward across the equatorial region, typically over a period of 30-60 days. The MJO is caused by the interaction between the atmosphere and the ocean, and is influenced by factors such as temperature, humidity,

and wind. During the active phase of the MJO, there is an increase in tropical rainfall, which can lead to flooding and other weather-related hazards. (Australian Government - Bureau of Meteorology, n.d.). Both the PNA and MJO and their effect on the weather is locked to their respective hemispheric regions, see Figure 5.

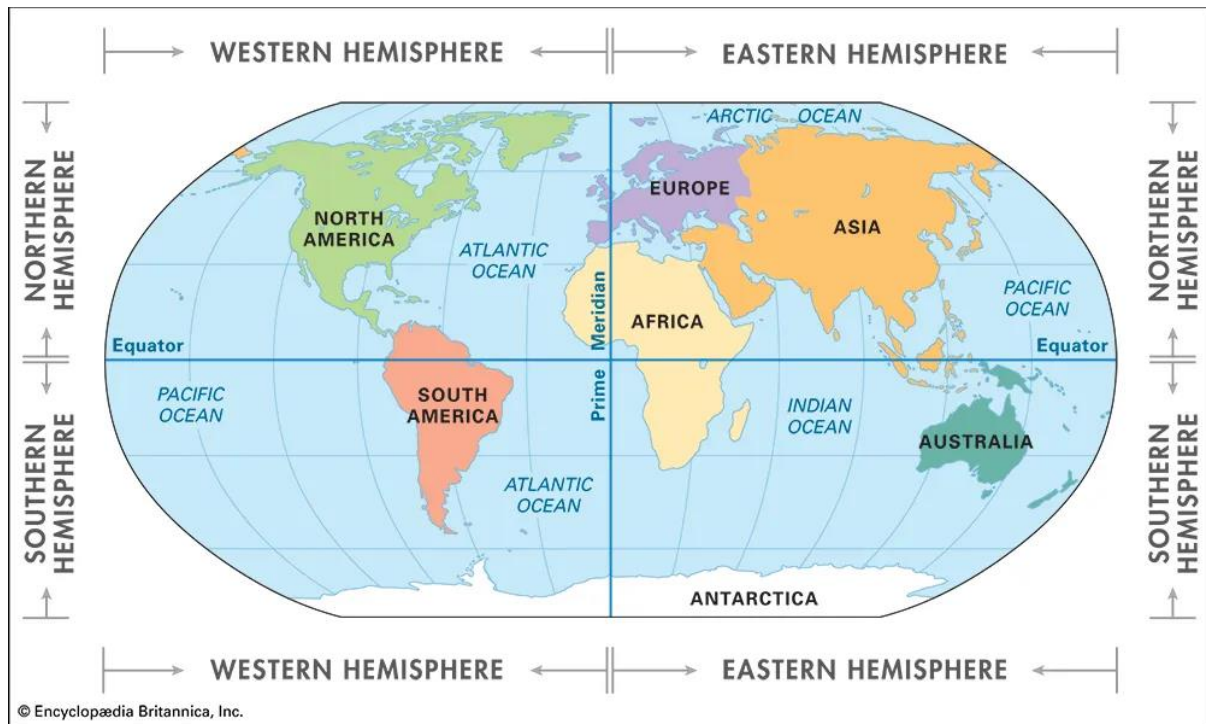


Figure 5: World map labelled with hemisphere and oceans (Encyclopædia Britannica, n.d.)

It was important to find climate phenomena which are regionally appropriate and affect SW England. Barnes et al. (2021) investigates the North Atlantic air pressure and temperature conditions related to heavy rainfall in Great Britain. Rain can be characterised by the synoptic scale weather conditions over the North Atlantic (see Figure 5). From 1950 to 2017, data were collected from 125 places throughout Britain. This data was then merged to show how pressure and temperature conditions alter to

create rainfall. It was found that when the NAO and AMO are in opposite phases there is heavy rainfall (Barnes et al., 2021) (see section 1.1.4). These two factors were used to predict rainfall as they fall into regions that affect the SW England: NAO has a 'strong effect on...weather in Europe' (Rafferty, n.d.) and changes of sea temperature AMO is taken over the North Atlantic Ocean (NCAR, n.d.). Its hemispheric relevance and the conclusion from Barnes is why these two meteorological parameters will be used to predict flash flooding in SW England.

2.4 Topography and catchments

Catchments with particular characteristics that flood quite easily are known as flashy catchments. In order to determine a flashy catchment, the water levels within the catchment get to maximum rise extremely quickly. To identify the variables that affect T_p , Langridge linked the catchment parameters of 153 gauges in the UK with 1400 storm event observations. Langridge used the regression formula supplied by the 1975 UK Flood Studies Report (FSR) FEH to calculate T_p (see equation 1 and figure 6).

$$T_p = 1.56 PROPWET^{-1.09} DPLBAR^{0.60} (1 + URBEXT_{1990})^{-3.34} DPSBAR^{-0.28} \quad (1)$$

$$T_p = 1.56 PROPWET^{-1.09} DPLBAR^{0.60} (1 + URBEXT_{1990})^{-3.34} DPSBAR^{-0.28}$$

where

PROPWET = Proportion of time when the soil moisture deficit (SMD) was less than or equal to 6 mm during the period 1961 to 1990 (Bayliss, 1999);

DPLBAR = FEH mean drainage path length index (km);

URBEXT₁₉₉₀ = Extent of urban and suburban land cover in 1990;
and,

DPSBAR = FEH mean drainage path slope index (m/km).

Figure 6: Abstract of peak time regression formula, annotated (Langridge et al., 2020; Kjeldsen, 2007)

2.5 Models in engineering

2.5.1 Numerical models

Mathematical models are key in any type of predictive endeavour. In its simplest form mathematical models take in data and simplify a problem to small systems of equations, algorithms, to predict future outcomes (Kidwell, 2022). Collier (2007) talks about the limitations of flash flood predictions and numerical models. Collier's academic journal presents the concept of numerical weather prediction (NWP), which uses data from the atmosphere and ocean to forecast the weather. Collier examined the limitations and unpredictability of forecasting techniques used for flash floods. NWP relies heavily on current, real-time information. Model results are very sensitive to changes in the starting circumstances (Thompson, 1957; Lorenz, 1963). This presents an issue: the initial-boundary-value problem, which is outside of the decided scope of

this project. The higher margins of error in most forecasting techniques are compensated for by the continuous stream of real-time data and slight changes in the environment. Data assimilation is crucial while calculating vulnerability in this project. That's why different types of data will be collected: topographical, meteorological, hydrological and time-series data will be obtained to lower the level of uncertainty when building this tool.

2.5.2 Machine learning algorithms and modelling

In recent years, the use of machine learning techniques has gained significant attention in the field of meteorology, particularly in predicting precipitation. By incorporating meteorological values as input variables into machine learning algorithms, it is possible to develop models that can accurately predict the occurrence of rainfall. The process involves feeding data into the algorithm, classifying the data, and using it to make predictions. The algorithm is then trained and tested on a set of data, and an appropriate model can be chosen based on its accuracy in predicting rainfall. The end result is a model that can be deployed and used to provide real-time predictions of rainfall. The ultimate goal is to develop a model that minimises prediction errors and provides accurate and reliable predictions of rainfall. Abrahamsen (2018) used machine learning to forecast the weather based on available weather data and used an application programming interface (API) to read meteorological data. Artificial neural networks (ANN) were developed using TensorFlow, a machine learning software. Holmstrom et al., 2015 explores using machine learning (ML) techniques for weather forecasting, his study focused on predicting maximum and minimum temperatures for seven days using weather data from the previous two days. The results showed that professional weather services outperformed both linear regression

and functional regression models, but the discrepancy between the models and the professionals diminished over time. The linear regression model performed better than the functional regression model, indicating that two days were too short to capture significant weather trends. Forecasting based on data from four or five days may allow the functional regression model to outperform the linear regression model. This academic paper proposes that with large amounts of time sensitive data ML models can provide accurate predictions. It also introduces the idea of creating multiple models and comparing the results which were used in creating this project's tool named SOPHIE.

CHAPTER THREE

METHODOLOGY

3.1 Research Methodology

3.1.1 Problem definition

This project aims to create SOPHIE, a tool which assesses the vulnerability of catchments to flash flood events. Vulnerability is going to be evaluated by the following two aspects: (1) determining whether the catchment is a flashy catchment and (2) whether the area experiences a flash flood event. If a flashy catchment is present, the area will fill quickly in response to rainfall making it easily susceptible to flooding. If both conditions are satisfied: the catchment is flashy, and rainfall is predicted to occur in the catchment region it can be concluded that the catchment is vulnerable to flash floods (See figure 7).

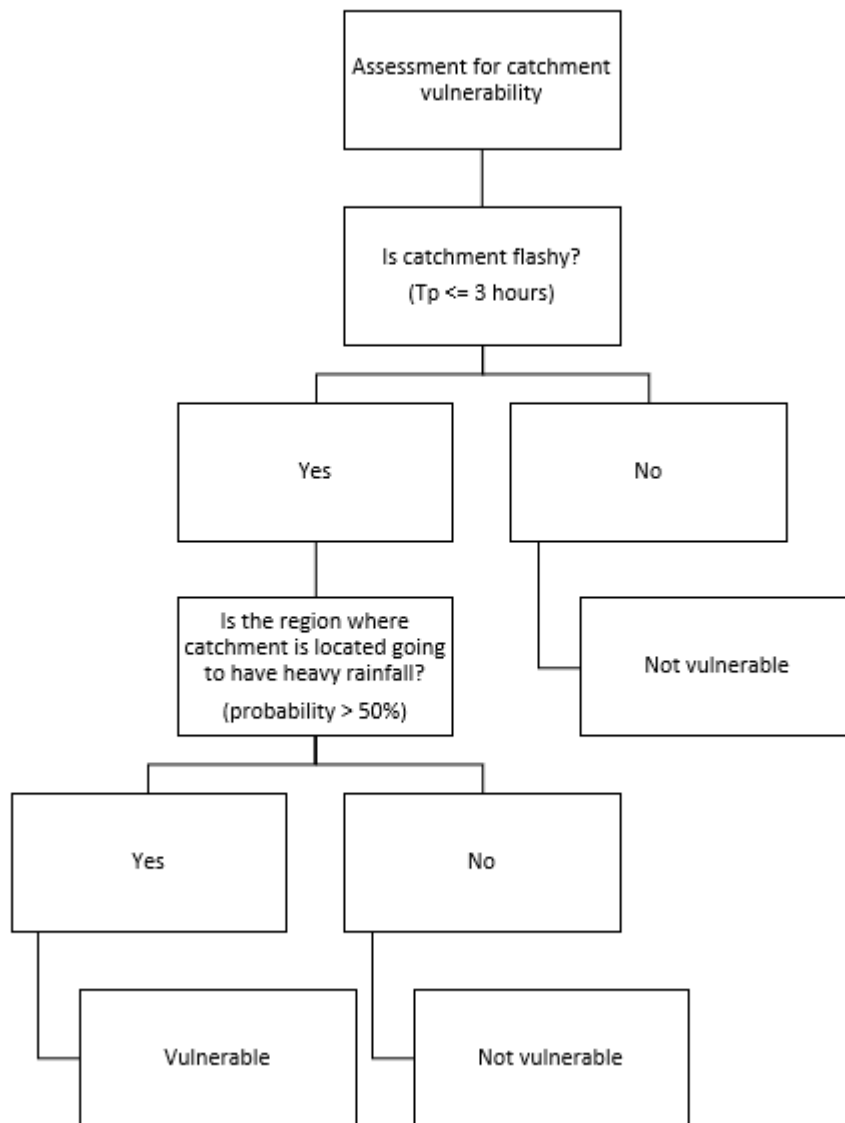


Figure 7: Flowchart to represent how vulnerability is assessed

3.1.2 Proposed solution and justifications

SOPHIE was developed using the programming language Python. It collects topographic information to determine the T_p for the user's chosen catchment. Next three different ML algorithms are used to predict the occurrence of a flash flood. As

mentioned in section 2.5.2 a model is a file which feeds data into an ML algorithm and makes a prediction based on the data. The dataset used for modelling was a collection of 181,637 non-flash flood and flash flood dates between 04/12/1960 to 22/11/2016. Most academic studies take into account topographic, hydrological, and meteorological factors independent of each other; however, SOPHIE includes important factors from each sector and offers a comprehensive evaluation of vulnerability. Figure 8 shows how SOPHIE will work.

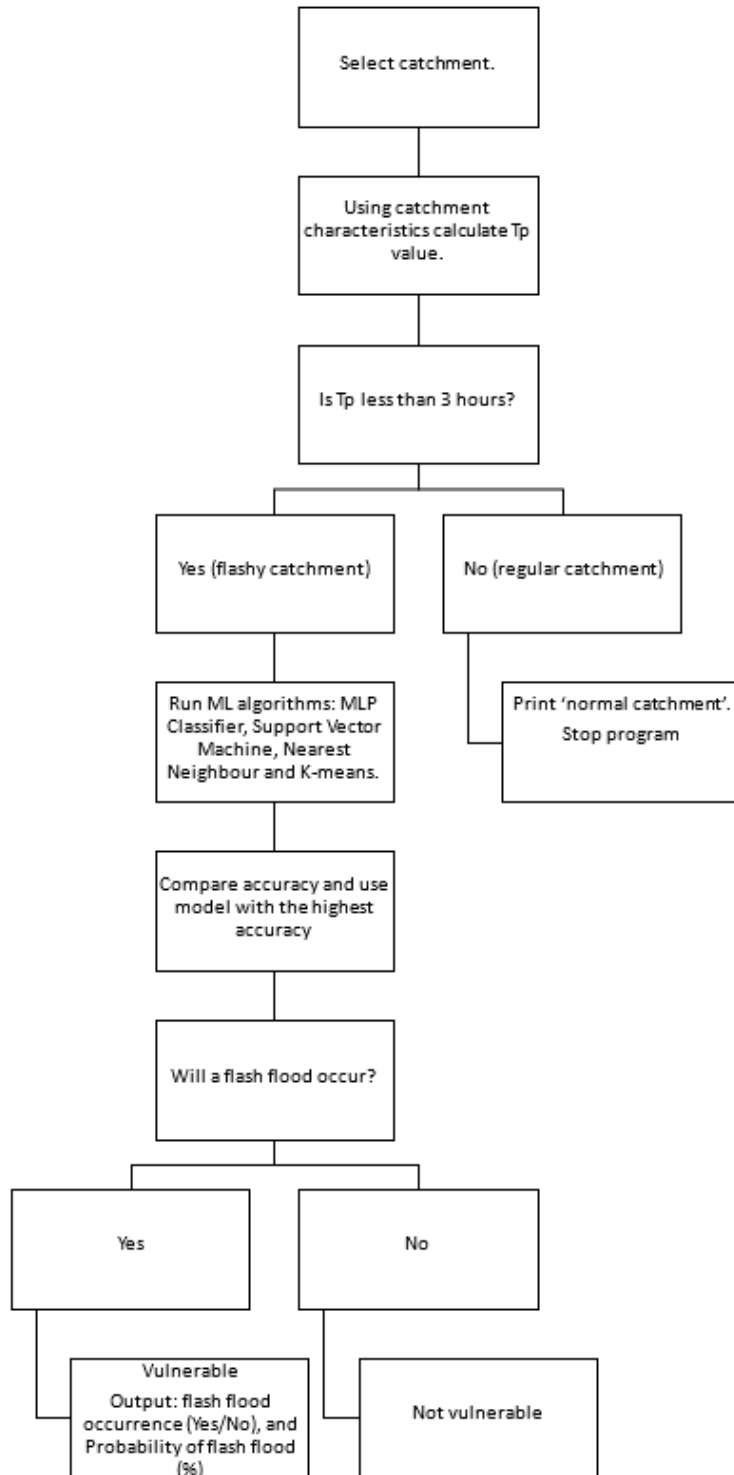


Figure 8: Flowchart displaying how SOPHIE measures vulnerability and expected outputs (See Appendix E-1)

3.2 Materials

This section discusses the software and hardware components required to run SOPHIE. Other essential material not mentioned include necessary data such as flash flood and non-flash flood dates for the chosen area and NAO/AMO index from a reliable source.

3.2.1 Software components

Requirements to run SOPHIE are the following: Python (3.11 to 3.8), Anaconda 3, Scikit-learn install (includes numpy and matplotlib library), Pandas library and Jupyter Notebook.

3.2.2 Hardware components

Minimum hardware components including modern operating systems for example Windows 7 to latest version, Mac OS X10.11 or higher, Linux:RHEL 6/7 (64-bit). 4 GB RAM and 5 GB free disk space (March, 2023).

3.3 Methods

A mixture of numerical and computational methods was used in the completion of this study, as SOPHIE includes multiple Python scripts which run calculations and algorithms to solve this engineering problem.

3.3.1 Data collection

Two datasets were needed in this project. (1) Dataset 1 is a compilation of catchment characteristics for all the catchment stations in SW England. (2) Dataset 2 includes all

flash flood events in the hydrometric SW region from 04/12/1960 to 22/11/2016 with their respective NAO/AMO numbers.

3.3.2 Dataset 1: Catchment station characteristics

Dataset 1 was provided by the Environment agency for research purposes. Each catchment station has a representative station number (see Appendix A-3).

3.3.3 Dataset 2: Southwest flash flood dates (pre-classification)

Next a dataset was created that contained 367 dates where flash floods occurred and 181,637 non-flash flood dates between 04/12/1960 to 22/11/2016. The main source was the British chronology of flash floods created by JBA Trust. JBA Trust is a charity which aims to increase understanding of water risk. They created the 'British Chronology of Flash Floods', a record for flood events in the UK using historical reports (JBA Trust Ltd, 2022). This was used in alignment with Archer's 'Historical flash floods in England: new regional chronologies and database' to create a new dataset which included the flash flood dates, month, year, NAO and AMO numbers (see Figure 9) and the non-flash flood dates separately with the same information (see Figure 10).

	Flashflood dates	Month	Year	NAO value	AMO value
0	09-Oct-59	10	1959	0.89	0.10
1	25-Nov-59	11	1959	0.41	0.10
2	22-Jun-60	6	1960	-0.21	0.09
3	23-Jun-60	6	1960	-0.21	0.09
4	01-Aug-60	8	1960	-1.40	0.08

Figure 9: Head of dataset 2 pre-classification, flash flood events

	Non flashflood dates	Month	Year	NAO number	AMO number
0	01-Oct-59	10.0	1959.0	0.89	0.1
1	02-Oct-59	10.0	1959.0	0.89	0.1
2	03-Oct-59	10.0	1959.0	0.89	0.1
3	04-Oct-59	10.0	1959.0	0.89	0.1
4	05-Oct-59	10.0	1959.0	0.89	0.1

Figure 10: Head of dataset 2 pre-classification, non-flash flood events

3.3.4 Dataset 2: Classified- uncompressed

The classification dataset 2 essentially cleans up the pre-classification dataset 2 by adding a classification element to the data, where 0 represents no flash flood and 1 flash flood. Also, the non-flash flood and flash flood dates were combined (see Figure 11).

3.3.5 Dataset 2: Classified- compressed

NAO/AMO values are given in a monthly and yearly format, for example the 10th of January 2016 and the 27th of January 2016 have the same NAO/AMO values. Dataset 2 was optimised and compressed by removing some days and only including one non-flash flood date per month. Compressing the data from 181,637 dates to 796 and therefore requiring less computer power to run the model.

	Year	Month	NAO	AMO	Flash Flood?
2580	12	1965	1.37	0.01	0
112	9	1974	0.82	0.17	1
8228	9	1981	-1.45	-0.16	0
13233	8	1995	0.69	-0.08	0
15397	9	2001	-0.65	0.08	0

Figure 11: Shuffled head of dataset 2 classified-uncompressed (see Appendix E-2)

3.4 Development SOPHIE

The development of SOPHIE can be split into two main sections the calculation of the T_p and the development of the ML model. See Figure 8 for the flowchart for SOPHIE.

3.4.1 Time to peak

File named 'Tp.py' was created to parse through the dataset 1. The user can input a catchment station number (see section 1.1.2) and the script will return the T_p value and whether the catchment is flashy or not (see Appendix A).

3.4.2 Correlation analysis and linear regression

Initially correlation analysis was used to analyse the data, which involves the graphing of the data and looking for any clear trends. Figures 12 and 13 are graphs which show the relationship between NAO and AMO for flash flood and non-flash flood dates respectively; where blue points represent NAO and orange represents AMO. This approach did not prove successful as the data was clustered and not linear due to the time-series element. An attempt was made to look at the data yearly, but no clear trend was evident which indicates the use of more complex modelling techniques.

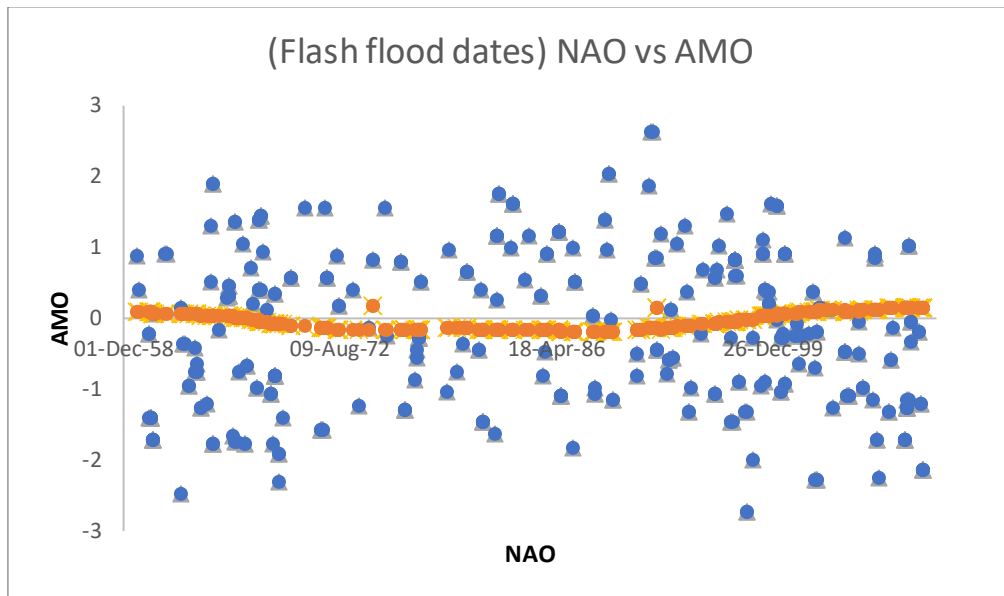


Figure 12: NAO (blue) vs AMO (orange) graph for all flash flood dates

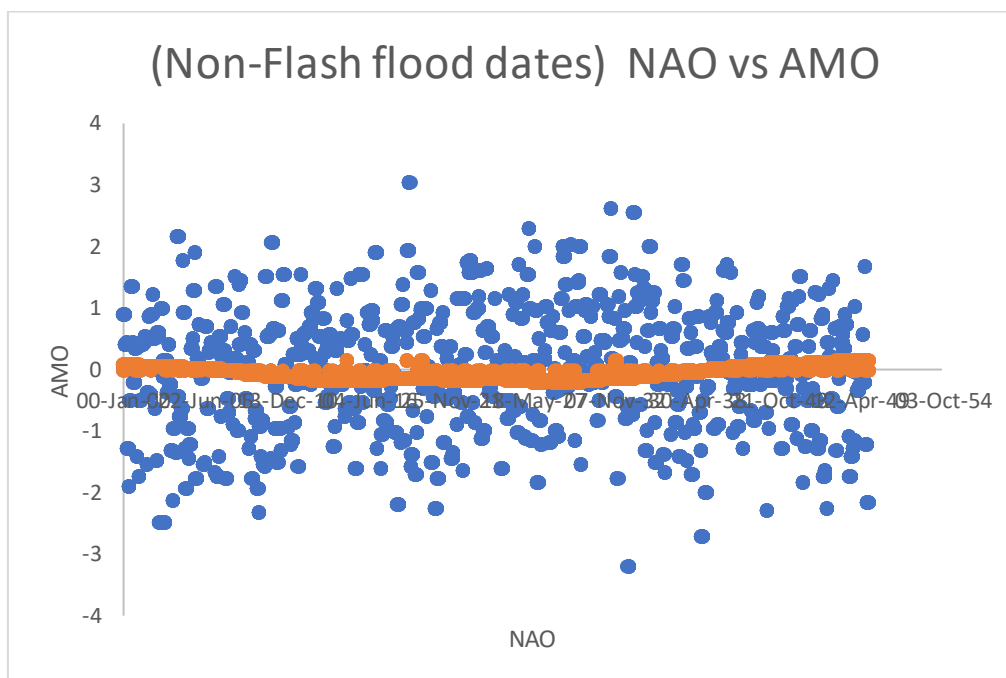


Figure 13: NAO (blue) vs AMO (orange) graph for all non-flash flood dates

The regression formulas were found (see Appendix B-1 and B-2). Equation 2 is the flash flood formula and equation 3 is the non-flash flood formula.

$$y = -0.0276x - 0.0311 \quad (2)$$

$$y = -0.0113x - 0.0547 \quad (3)$$

where

$x = NAO$

$y = AMO$

However, this approach was too simplistic and did not work because of the non-linear relationship between the variables. Correlation analysis and regression assumes a linear relationship. The regression equation only helps to find the NAO given the AMO (or vice versa) in an assumed flash flood or non-flash flood event which is not useful. Although the graphs served as good data visualisation tools it was clear that more complex models were needed and data needed to be changed to a classification dataset, where a classification variable could be used to aid prediction.

3.4.3 Machine Learning algorithms

Machine Learning (ML) is a type of artificial intelligence that involves feeding data into an algorithm to improve outcomes through experience. The process typically involves the use of classification and prediction algorithms. Prior to using ML in this project, it was important to acquire and clean up the data that was going to be used (see section 3.3.4). The data was then separated into a training set (80%) and a testing set (20%). The training data was used to build a model, which was then validated for accuracy using the testing data. Once multiple algorithms were tested and the one with the highest accuracy was chosen, it could be used to make predictions on new data.

Three ML models are built using the algorithms below:

1. Neural network Multilayer Perceptron (MLP) Classifier
2. Support vector machine (SVM)
3. K-Nearest neighbour (KNN)

3.4.3.1 MLP Classifier

A MLP Classifier is a type of neural network that is commonly used for classification tasks (Brownlee, 2022). It consists of multiple layers of interconnected nodes, where each node is a simple computational unit that takes input from the previous layer and produces an output that is passed to the next layer (see Figure 14). The input layer of the MLP takes in the features of the data to be classified, while the output layer produces the predicted class labels. The hidden layers in between the input and output layers are where the majority of the computation takes place. During training, the weights, and biases of the MLP are adjusted to minimise the error between the predicted class labels and the true class labels of the training data. Once the MLP has been trained, it can be used to classify new data by passing the features through the network and producing a predicted class label based on the output of the final layer (scikit-learn, n.d.).

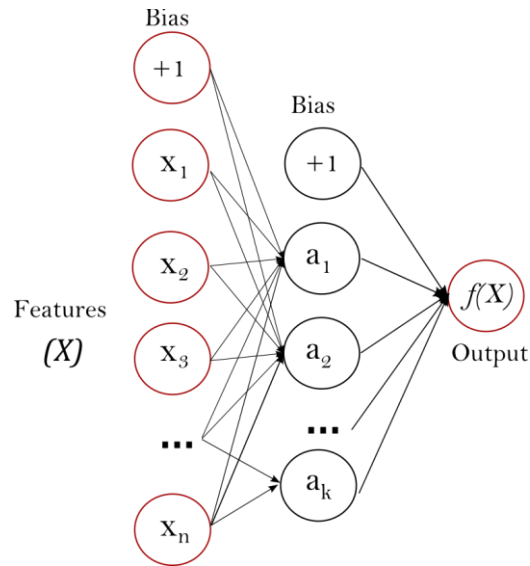


Figure 14: MLP Classifier diagram, one hidden layer (scikit-learn, n.d.)

3.4.3.2 Support Vector Machine

A SVM is a type of machine learning that separates data into different classes using a line or plane called a hyperplane. The SVM looks for the hyperplane that separates the two classes with the largest margin in binary classification. The SVM can transform data into a higher-dimensional space using the kernel trick to capture more complex relationships (scikit-learn, n.d.). During training, the SVM adjusts the hyperplane's parameters to minimise the error between the predicted and true class labels of the training data using a method called gradient descent. This allows the SVM to classify new data points as belonging to one class or the other based on which side of the hyperplane they fall on (Suthaharan, 2016).

3.4.3.3 K-Nearest Neighbour algorithm

The kNN algorithm is a machine learning method used for classification and regression tasks. The algorithm works by identifying the k closest data points in the training set to

a new input and using their known labels or values to predict the label or value of the new input (IBM, n.d.). The 'k' in KNN is a parameter that refers to the number of nearest neighbours or classification groups in this project k would be 2. One group for flash floods and another for non-flash floods (Subramanian, 2019).

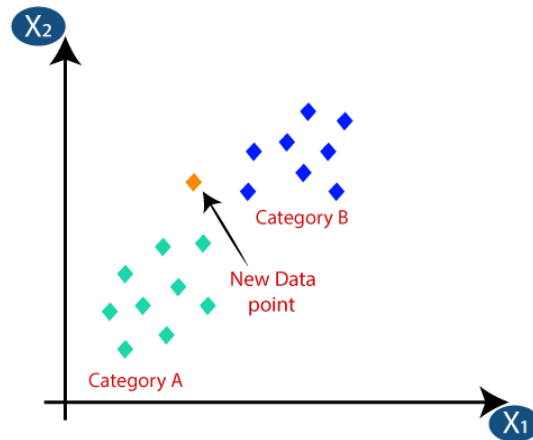


Figure 15: kNN diagram where $k=2$ (javaTpoint, n.d.)

3.4.3.4 Comparing ML Algorithms

3.4.3.4.1 SVM vs kNN

Hamilton et al., 2020 compares Support Vector Machines (SVM) and k-Nearest Neighbour (kNN) machine learning algorithms for classifying images. SVM is faster and more accurate than kNN in classifying images, occasionally misclassifying large objects but generating few small misclassifications that would interfere with the final classified image. Both algorithms yield positive results, but SVM provides significantly better classification accuracy and speed than kNN (Hamilton et al., 2020).

3.4.3.4.2 MLP vs SVM

Thomas, 2019 discusses a comparison between Support Vector Machine (SVM) and Multilayer Perceptron (MLP) machine learning models, using a Wisconsin Breast Cancer Dataset to predict if a patient has a benign or malignant cancer type. Thomas concludes that SVM's typically have better accuracy than MLP (Thomas, 2019).

3.4.3.4.3 kNN vs MLP

Eskandarin et al., 2010 researched into MLP Neural Network to predict the flow of Bakhtiari River, they also used the kNN method. They compared the performance of both models. The MLP model performed slightly better than the kNN method (Eskandarin et al., 2010).

Based on the research, it can be hypothesised that SVM is the most accurate algorithm, followed by MLP and then kNN.

CHAPTER FOUR

RESULTS AND DISCUSSIONS

4.1 Results

This section presents the outputs from the different algorithms tested, the algorithms were changed and tested to explore accuracy. The section is divided into two main parts: testing the model on three case studies (two flash floods and one non-flash flood event) and accuracy of the algorithms.

4.1.1 Case studies

This section explores the validity of the models by analysing data from two flash floods and one non-flash flood event, to identify if the model can predict a flash flood event when the outcome is known. Table 2 shows the chosen events.

Table 2: Case study information

Location	Catchment station (closest)	Date	Flash flood (Yes/No)
Boscastle	49006	16/08/2004	Yes
Totterdown (Bristol)	53017	28/07/2008	Yes
Stour at Throop	43007	28/01/2002	No

Outputs from the models are displayed in their respective sub-sections. How vulnerability is assessed based on SOPHIE's outputs are shown in Appendix E.

4.1.1.1 Flash flood event: Boscastle, 2004

The outputs for Boscastle are as displayed in Table 3, this catchment is determined not vulnerable, as the catchment is flashy, and the probability of a flash flood occurrence is less than 50%.

Table 3: SOPHIE outputs: Boscastle 2004 (49006)

Flashy catchment (Yes/No)	Yes
Time to peak value	2 hours 27 minutes
Flash flood occurrence (Yes/No)	No
Probability of flash flood occurrence	1%(MLP), 2%(SVM)
Vulnerability	Not vulnerable

4.1.1.2 Flash flood event: Totterdown, 2008

The outputs for Totterdown are as displayed in Table 4, this catchment is determined not vulnerable, as the catchment is not flashy, so the SOPHIE did not run the ML model.

Table 4: SOPHIE outputs: Totterdown 2008 (53017)

Flashy catchment (Yes/No)	No
Time to peak value	5 hours
Flash flood occurrence (Yes/No)	No
Probability of flash flood occurrence	4%(MLP), 2%(SVM) (manual)
Vulnerability	Not vulnerable

4.1.1.3 Non-flash Flood: River Stour, Throop, 2002

The outputs for Throop are as displayed in Table 5, this catchment is determined not vulnerable, as the catchment is flashy, and the probability of a flash flood occurrence is less than 50%.

Table 5: SOPHIE outputs: Throop 2002 (43007)

Flashy catchment (Yes/No)	No
Time to peak value	18 hours 7 minutes
Flash flood occurrence (Yes/No)	No
Probability of flash flood occurrence	2%(MLP), 2%(SVM)
Vulnerability	Not vulnerable

4.1.2 Summary of Results

SOPHIE predicted one out of the two case studies correctly, Throop 2002 not vulnerable. However, SOPHIE predicted Boscastle 2004 and Totterdown 2008 incorrectly, both as not vulnerable despite both being confirmed flash flood events and therefore vulnerable catchment regions.

4.1.3 Model accuracy

Scikit-Learn is a Python library used to implement ML models and was used to develop SOPHIE. One of scikit-learn's features includes an accuracy score function which uses the testing data to verify the model's accuracy (See Figure 16).

```
accuracy = accuracy_score(y_test, y_pred)
print('Accuracy:', accuracy)
```

Figure 16: Accuracy scorer function (top line)

Each model was tested using the accuracy scorer function. The findings are presented in Table 6. To increase the accuracy each algorithm had a slight variation. For the MLP classifier the number of layers was increased by one which adds more neurons to the network allowing the classifier to learn more, the SVM remained the same as the classifier is a built-in function and changing its dependencies falls outside of the scope of this project and for kNN, the k parameter dictating the number of classification groups was increased by one (see section 3.4.3). From the following results shown in Figure 17. The following algorithms, variation and dataset were chosen for the vulnerability model: MLP with hidden layers 10,10,10 and SVM both pulling data from dataset 2 classified-uncompressed.

Table 6: Accuracy results for each model

Model	Dataset	Variation	Accuracy (decimals)	Accuracy (%)
MLP	Dataset 2 (compressed)	'hidden_layer_sizes=(10,10)	0.6415094339622641	64.151
		'hidden_layer_sizes=(10,10,10)	0.6729559748427673	67.296
	Dataset 2 (uncompressed)	'hidden_layer_sizes=(10,10)	0.9796423658872077	97.964
		'hidden_layer_sizes=(10,10,10)	0.9796423658872077	97.964
SVM	Dataset 2 (compressed)	(no variation)	0.6352201257861635	63.522
	Dataset 2 (uncompressed)	(no variation)	0.9796423658872077	97.964
kNN	Dataset 2 (compressed)	'n_neighbors=2'	0.7238493723849372	72.385
		'n_neighbors=3'	0.6443514644351465	64.435
	Dataset 2 (uncompressed)	'n_neighbors=2'	0.9776228906823184	97.762
		'n_neighbors=3'	0.9776228906823184	97.762

Table 6 is represented in a graph in figure 17. The blue bars are dataset 2 classified-compressed and the red bars are dataset 2 classified-uncompressed.

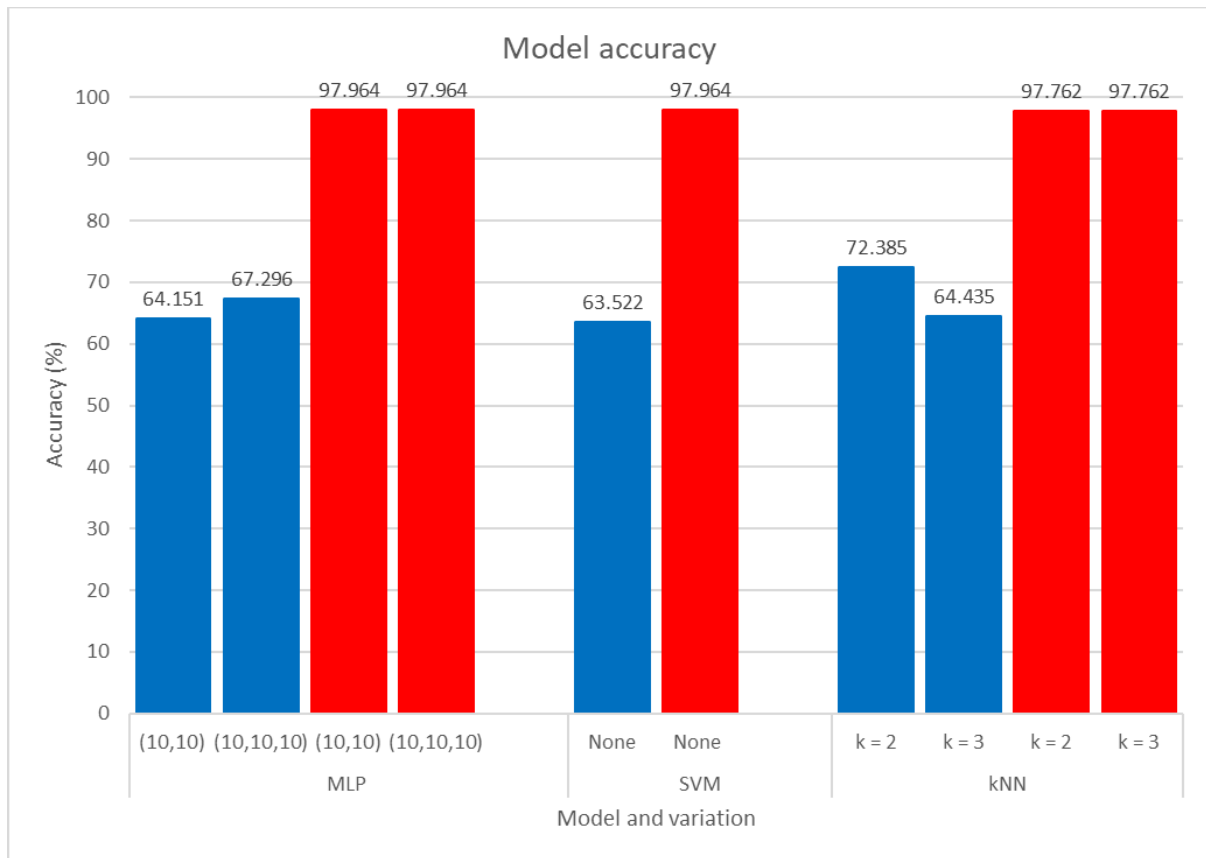


Figure 17: Graph of accuracy for each model

4.2 Discussion

The main project aim was met. A tool which assesses flash flood vulnerability in the SW was developed. This was a multidisciplinary exploration where data from three different disciplines were successfully implemented. (1) Topography by using information about the catchment stations to determine whether a catchment was flashy. (2) Hydrology by investigating and creating a dataset of past flash flood events in SW England. (3) Meteorology by building models to predict flash flooding using the weather parameters: NAO/AMO. The secondary aim of this project also met, although the model failed twice and succeeded once when assessed against three case studies

it established that NAO/AMO values are not the best parameters to use for flash flood prediction.

However, all of the project objectives were met. The tool, SOPHIE was developed as previously stated. Three machine learning algorithms were tested and compared, and the most accurate algorithm was used to predict the probability of flooding, which is catchment vulnerability to flash flooding. Furthermore, insightful conclusions were made based on the findings.

On 16th August 2004 a flash flood occurred in Boscastle. Heavy rainstorms caused two rivers to overflow near the village of Boscastle, resulting in approximately two billion litres of water rushing down the valley and into Boscastle (Met Office, n.d.). This led to severe damage to buildings and cars. The incident caused millions of pounds worth of damage (Met Office, n.d.). This flash flood event was tested using the model built. The nearest catchment station is 49006 which has rivers that flow into the Boscastle. The Met Office reported that 75mm of water fell in the space of 2 hours and states this is the same amount that falls throughout the entirety of August, which is close to the T_p . (Met Office, n.d.). This matches the output T_p shown in Table 3: 2 hours and 27 minutes. SOPHIE did not accurately predict the expected flash flood outcome. This outcome represents a limitation on the efficacy of the machine learning model, specifically with respect to the input and parameterization employed (NAO/AMO).

A lesser-known flash flood occurred on 28th July 2008 in Bristol, catchment station 53017 was inputted through SOPHIE found the catchment to be a normal catchment.

However as discussed in section 2.2 water levels can rise and flooding can occur despite having normal catchment characteristics. SOPHIE did not run after this check (see Figure 8). However, the NAO and AMO values were manually input to see if SOPHIE would return a correct flash flood prediction, which it did not. Vulnerability in this project is defined as first a catchment being a flashy catchment and then rainfall with a T_b of six hours occurring in the catchment region. This case study did not provide expected results but also introduces an interesting limitation to how vulnerability is being assessed in this project.

At random a catchment station and date were selected. To ensure no flash floods had occurred both date and location were checked against dataset 2, JBA Trust's flash flood chronology and Met Office online public databases. The model tested catchment station 43007 (Stour at Throop) on 28th January 2002, which is a non-flash flood date. SOPHIE outputted the catchment as normal and predicted there would be no flash flood making the catchment safe, which is correct.

From the results of the case studies multiple conclusions can be made. Research conducted by Barnes et al., 2021 suggested that when the NAO and AMO are in opposite phases there is heavy rainfall. This theory was used to model flash floods as it suggested a correlation between NAO/AMO and rainfall. However heavy rainfall and flash floods are different as they have different T_b . Therefore, while ML models may be able to predict heavy rainfall based on the phases of the NAO and AMO, they may not be able to accurately predict flash floods, which may require a different set of variables to be taken into account. The AMO and NAO are climate indices that describe

variations in sea surface temperatures and atmospheric pressure patterns over the Atlantic Ocean. While they can be used to understand long-term climate patterns, they are not typically used to directly predict flash floods; this introduces a limitation which will be referred to as differences in scales. This is due to NAO/AMO indices being great predictors of rainfall on a large geographical or regional scale (North Hemisphere) (see Figure 5); however, flash floods are extremely small localised events that do not affect an entire hemispheric region. Flash floods are typically caused by intense precipitation events, which are difficult to predict accurately more than a few days in advance (Champion, 2014). While climate patterns can influence the likelihood of extreme precipitation events, they are just one of many factors that can contribute to the development of a flash flood.

The shortcomings observed in the case studies could be due to data limitations and inaccuracies. The data used was static which is ideal for problems that do not change over time. That is why the outputs regarding the T_p values were correct all the time as the T_p is calculated by the measurements taken from catchment stations, which for the most part, goes unchanged. However, the NAO/AMO climate indices introduce a time series element into the model. In the literature review there was criticism of the NWP models overreliance on a feed of real-time information (see section 2.5.1); unironically the lack of real-time data and time restriction could be one of the limiting factors for the models tested.

In order to make up for the lack of access to real-time meteorological data, the decision to use the uncompressed version dataset 2 was made, so there was more training data available. The use of uncompressed data may result in misleading accuracy, particularly in the context of binary classification problems involving large datasets. The task of dividing the data into two groups is known as binary classification (Mishra, 2021); in this instance no flash flood and flash flood. As an assumption can be made that each datum belongs to the classification group with the majority leading to artificially high accuracy scores despite potential errors in classification.

Additionally, man-made errors in the dataset, such as missing or incorrect data, can further reduce the performance of SOPHIE. While multiple reputable sources were used to negate this, it is possible the models were trained on data with outliers and inaccuracies which could in turn affect the output. Therefore, it is necessary to acknowledge this possibility when evaluating its performance.

CHAPTER FIVE

CONCLUSIONS AND FUTURE WORK

5.1 Key findings

5.1.1 Catchment characteristics can be used to calculate T_p

The key findings from the project indicate that catchment station measurements can be used to calculate the T_p of a catchment. Specifically, a catchment with a T_p of less than three hours can be classified as a flashy catchment. This finding has significant implications for the project, as identifying flashy catchments was a necessary condition for the model to consider meteorological parameters. In essence, the project highlights the importance of catchment station measurements in characterising the hydrological behaviour of catchments and its relevance in building hydrological models.

5.1.2 MLP is the most accurate algorithm for binary classification

The key findings of the project reveal that MLP classification is more accurate than SVM and kNN algorithms for binary classification. Even though MLP and SVM had the same accuracy score for dataset 2 classified-uncompressed, the MLP with hidden layers 10,10,10 performed better in dataset 2 classified-compressed, contrary to the hypothesis proposed at the end of section 3.4.3.4.3. The impact of this discovery on the project was significant as SOPHIE selected the most accurate algorithm based on the accuracy scorer function once the T_p condition was met. On a larger scale, these findings can have implications for modelling and machine learning in various industries. MLP's superior performance for binary classification may lead to more widespread use of this algorithm in fields that require accurate classification. These results also suggest

that researchers should continue to explore and compare different algorithms to determine which is best suited for specific classification tasks.

5.1.3 Predictive models need real-time data

SOPHIE has data limitations, and to increase its accuracy, a larger dataset is needed. Additionally, real-time data updates on catchment characteristics from NRFA and reliable sources on NAO and AMO are required. This would have a significant impact on the project, as a larger and up-to-date dataset would result in a more accurate model. Real-time data is particularly relevant in modelling, especially predictive rainfall models, as it allows for more accurate predictions. Therefore, it is crucial to incorporate real-time data into modelling tools to improve their accuracy and usefulness.

5.1.4 Redefinition of how vulnerability is evaluated

The evaluation of vulnerability is crucial to how SOPHIE works. The tool has two stages, the flashy catchment verification, and whether the area is going to rain. If the catchment is not flashy, the tool does not progress to the second stage and assumes the catchment is safe. This presents a major limitation and over reliance on how catchments are determined flashy or not. SOPHIE uses the equation 1 the peak time regression formula (Langridge et al., 2020; Kjeldsen, 2007), which is extremely sensitive to certain catchment characteristics (see Figure 6). Using the T_p to determine if a catchment is flashy needs to be evaluated, tested, and challenged. Additionally, water levels can still rise in non-flashy catchments, an idea explored and backed with examples in section 2.2. This finding has a significant impact on the project as it aids

in explaining potential errors in SOPHIE's outputs. As SOPHIE does run unless the condition of a flash catchment is first met. Furthermore, it has broader implications for the use of conditions in hydrology models. Rethinking conditions is essential as their misapplication can cause significant inaccuracies, leading to incorrect predictions and decisions.

5.1.5 NAO/AMO are linked to heavy rainfall

The NAO and AMO have a correlation with heavy rainfall rather than flashy floods. Heavy rainfall can have a total duration of well over 6 hours, which is longer than what is typically considered a flash flood. This finding has a significant impact on the project, as it highlights the inaccuracy in the models that classify data into no heavy rainfall and heavy rainfall days. Modelling the model after the wrong parameters can lead to incorrect predictions. Therefore, it may be necessary to find more appropriate parameters and conduct further research to improve the accuracy of the models. This finding also has broader implications for modelling, as it emphasises the importance of selecting appropriate parameters to build a model. It highlights the need for caution in selecting and interpreting data to ensure the models' accuracy and usefulness in decision-making.

5.2 Conclusions

In conclusion, the study has shown that accurate weather predictions which are pivotal for vulnerability assessment and safety measurement tools require real-time data, and the NAO/AMO relationship can be useful for predicting heavy rainfall but not flash

floods due to the differences in scales. However, evaluating the vulnerability of areas to flash floods would require a different infrastructure for the tool SOPHIE, as the initial topographical condition limits SOPHIE's accuracy. Additionally, the comparison of classification algorithms showed that MLP was the best option for binary classification tasks. These findings have important implications for the meteorological industry and highlight the need for continued research and development in this field to improve weather prediction accuracy and better protect communities from the impacts of extreme weather events seeing the worsening of climate change.

5.3 Future Work

The future work for this project would involve significant changes to SOPHIE's infrastructure, namely the evaluation of vulnerability, as well as researching other climate indexes and relevant climate parameters to model specifically flash floods which resolves the differences in scale issue highlighted in section 4.2. Lastly researching with a focus on calculating T_p and how flashy catchments are defined needs to be conducted. Additionally, implementing real-time data through APIs and creating a user-friendly version of the tool using Flask would be essential. As of this moment, SOPHIE serves as the back-end code, thus it's necessary to develop certain front-end code features to enable other researchers to use the tool similar to Archer's online database shown in Figure 3. As the project is developed using Python, this can be achieved using Flask. Flask is a 'web framework' which allows web applications to be created from the Python language (Dyourri, 2021). Also, real-time data could be implemented through the use of APIs which was not done in the completion of this project. These improvements would help to make the tool more useful for flood

management and improve its practical relevance. To further improve the project's impact, it may be necessary to conduct additional research on the correlations between climate indexes, meteorological parameters, and flash floods. Additionally, involving third party members such as flood management agencies and local communities in the tool's development and implementation could lead to more practical and effective use of the tool. The findings of this project may be useful in tackling the increasing threat posed by flash floods, a phenomenon strongly linked to climate change. Future research and improvements could yield significant benefits on flood management strategies and decision-making.

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APPENDICES

Appendix A

Appendix A-1: Time to peak Python script (Tp.py)





Appendix A-2: Tp.py terminal output

```
Enter catchment station number in Southwest England: 43007
Time to peak value for this catchment is: 18 h 7 m 0 s
Normal catchment
```

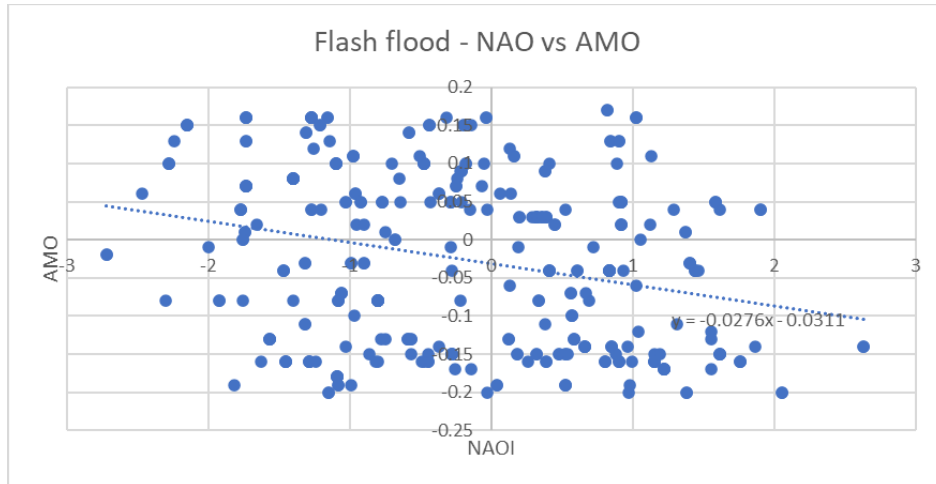
Appendix A-3: condensed head of Dataset 1

Station	River	Location	hyd_area	Map	Ireland	SCORE	prop.summer	Tp
43003	Avon	East Mills Total	43	9	False	4	0.062500	16.760089
43004	Bourne	Laverstock	43	9	False	4	0.088889	11.304268
43005	Avon	Amesbury	43	9	False	3	0.100000	11.477511
43006	Nadder	Wilton	43	9	False	3	0.076923	9.035584
43007	Stour	Throop	43	9	False	5	0.108696	18.126149

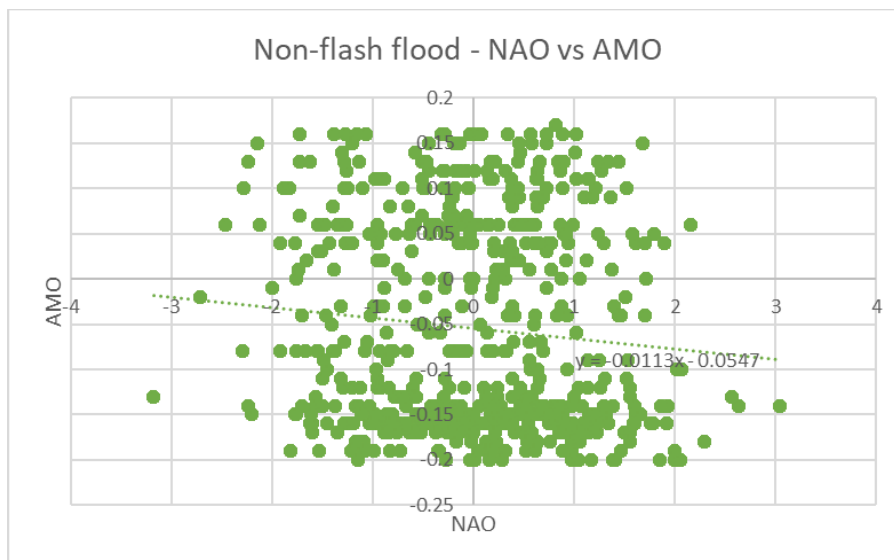
**See row five*

Appendix B

Appendix B-1: NAO vs AMO flash flood graph with regression formula

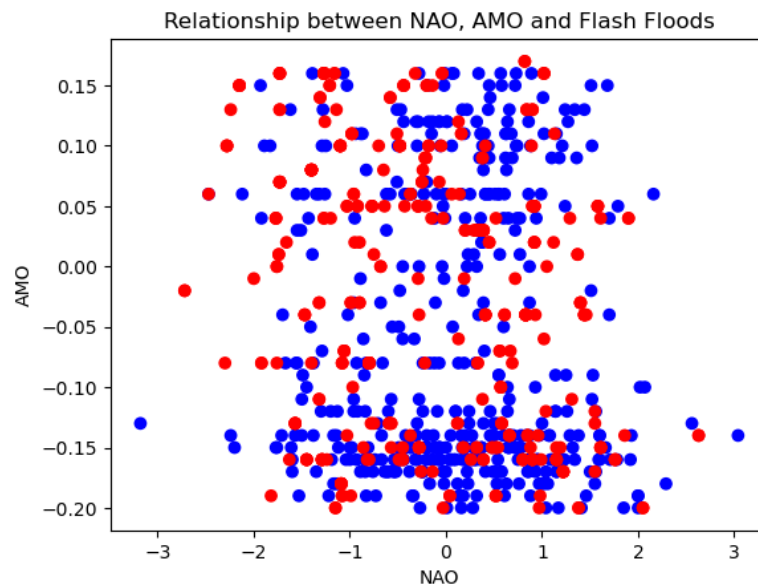


Appendix B-2: NAO vs AMO non-flash flood graph with regression formula

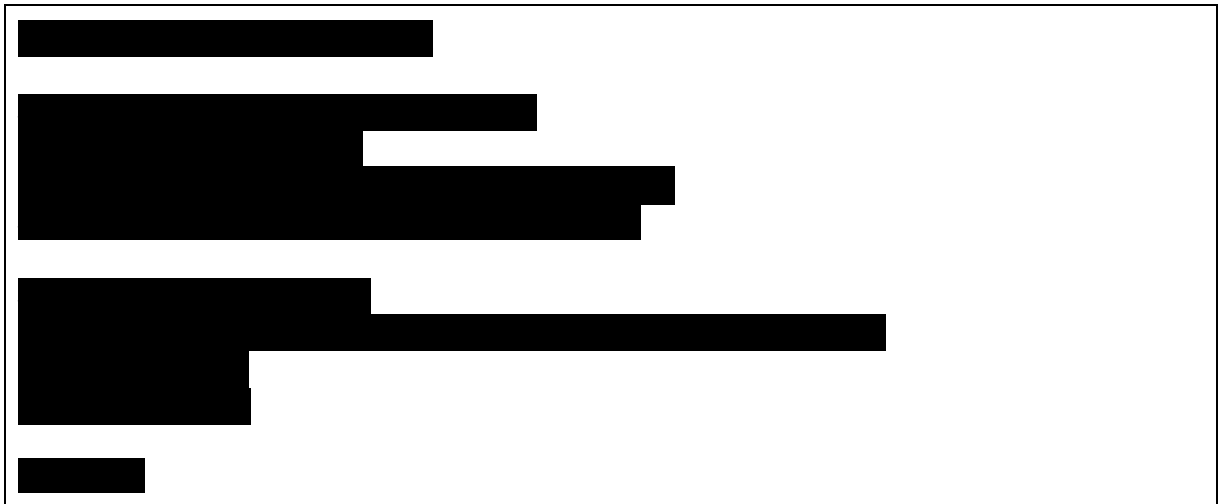


Appendix C

Appendix C-1: NAO vs AMO graph (dataset 2 classified-compressed)



Appendix C-2: Code that produced Appendix C-1



Appendix D

Appendix D-1: source code for kNN classifier used in SOPHIE

```
[REDACTED]
```


[REDACTED]

Appendix D-2: source code for MLP classifier used in SOPHIE

[REDACTED]

[REDACTED]

██████████

[REDACTED]

██████████

[REDACTED]

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Appendix D-3: source code for SVM classifier used in SOPHIE

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Appendix E

Appendix E-1: Key for probability of flash flood occurrence and vulnerability

SOPHIE output - probability of flash flood occurrence	Vulnerability outcome
49% or less	Not vulnerable
50% or more	Vulnerable

Appendix E-2: Code for Figure 11 – shuffled head of data frame

[REDACTED]

[REDACTED]

[REDACTED]

Appendix F

Appendix F-1: Gantt Chart Table

Task Name	Duration	Start	Finish
Project Specification and Literature Review	32	26/09/2022	28/10/2022
Select dissertation topic	4	26/09/2022	30/09/2022
Read academic journals	7	01/10/2022	08/10/2022
Take notes on literature	6	09/10/2022	15/10/2022
Write up specification	8	16/10/2022	24/10/2022
Submit report	3	25/10/2022	28/10/2022
Receive Feedback	0	01/11/2022	01/11/2022
Programming the tool	119	02/11/2022	01/03/2023
Dataset curation - flash floods/ non flash flood dates with NAO and AMO values	28	02/11/2022	30/11/2022
Create script calculates Tp values based on catchment station	14	01/12/2022	15/12/2022
Experiment with different machine learning models	56	16/12/2022	10/02/2023
Select best model	2	10/02/2023	12/02/2023
Constructionarium (academic break)	5	12/02/2023	17/02/2023
Train model	10	18/02/2023	28/02/2023
Project Report	30	01/03/2023	31/03/2023
Write up project	30	01/03/2023	31/03/2023
Submit report	1	30/03/2023	31/03/2023
Poster Presentation Viva with Employability Document	53	31/03/2023	23/05/2023
Produce a poster presenting work + Gather notes for logbook	19	01/04/2023	20/04/2023
Write an employability document	24	21/04/2023	15/05/2023
Attend the presentation	0	23/05/2023	23/05/2023

Appendix F-2: Gantt Chart

